

A Per-Instrument Meta-Model for a Primary Trading Signal

*Triple-Barrier Labels, CPCV(6,2) Champions, and a
Sharpe-Optimal Sizer across Equity, Energy and Metals Futures*

Universe

11 instruments
3 asset classes
2020-01 to 2022-06

Deliverables

`metamodel_predictions.csv`
`strategy_weights.csv`
1,408 rows each (128 days $\times$ 11)

Authors

Sreeram Andra · Harry Browne · Sacha Mimoun
Jay Jung · Aiden Berendes

Branch: `submission/harden` • Date: June 2026
Reproducibility seed: `random_state = 42`

Executive Summary

This report documents a per-instrument meta-labelling pipeline that takes the supplied primary signal $s_{i,t} \in \{-1, 0, +1\}$ for 11 futures contracts (three equity indices, four energy contracts and four metals contracts) and predicts, for each non-zero signal day, the probability that following that signal would have been profitable under a triple-barrier exit. The optional strategy track converts the calibrated probabilities into vol-targeted position weights.

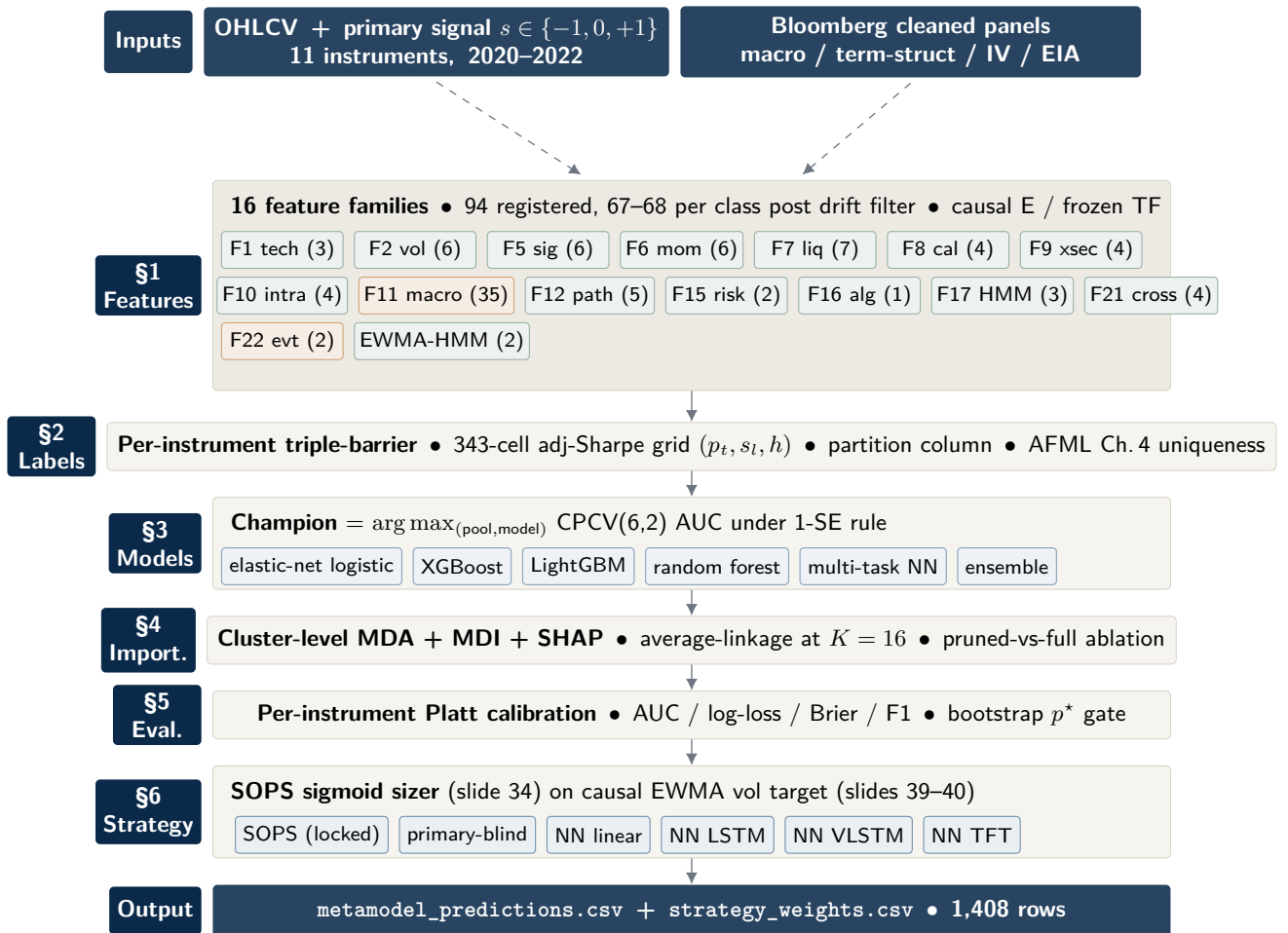
The pipeline is built around four design commitments. First, label geometry is selected *per instrument* from a 343-cell grid $(p_t, s_l) \in \{0.25, \dots, 2.5\}^2 \times h \in \{1, \dots, 20\}$ by an adjusted-Sharpe objective on the development partition only, producing the geometries reported in Table 3. Second, every modelling decision sits behind a fully *purged, embargoed* cross-validation scheme (CPCV(6,2), 15 paths) following López de Prado (2018) Ch. 7 and 12, with per-instrument embargo widths derived from each instrument’s own label-span distribution (`embargo_p90` in `instrument_scope.json`). Third, model selection uses a *one-standard-error rule* over a four-candidate roster (elastic-net logistic, XGBoost, LightGBM, random forest, plus a multi-task neural network) crossed with three candidate *pool* structures (individual, sub-class, full asset class). Fourth, the strategy layer implements the lecture’s Optional Session 3 Sharpe-Optimal Position Sizing (SOPS) sigmoid (slide 34) on causal EWMA-vol weights (slides 39–40), and benchmarks it head-to-head against four neural-portfolio backbones (Linear, LSTM, VLSTM, Temporal Fusion Transformer) trained with the Sharpe loss (slide 42).

The submitted deliverable is the SOPS sizer on the equal-weight ensemble of per-instrument champions. On the sealed H1 2022 test window (129 trading days, 1,002 events) the strategy realises an annualised Sharpe of **3.15** net of Grinold–Kahn transaction costs (2 bps fixed plus 10 bps proportional), $t = 2.25$ (per-period $t = \text{SR}\sqrt{n}$), a studentised stationary block-bootstrap 95 % confidence interval of $[0.022, 0.356]$ per period that *excludes zero*, a $\text{PSR}(0) = 0.989$ (above the deployment threshold of 0.95), and a Deflated Sharpe Ratio of 0.943 at $4 \cdot N_{\text{raw}} = 480$ candidate trials. Pesaran–Timmermann directional skill registers $S = 4.60$, $p < 10^{-5}$. Realised annualised volatility is 5.1 %, well below the 10 % cap.

The model has localised strengths and honest limitations. The per-class pooled CPCV(6,2) random-forest AUC benchmarks (with Bloomberg features) land at **equity 0.550, energy 0.670, metals 0.538**. The energy result is driven by a microstructure + regime cluster (`f5_participation_60`, `f7_oi_level`, `ewma_hmm_prob_highvol`) whose cluster-level MDA reads +0.129, an order of magnitude above any other cluster in any class. The equity panel admits one MDA-significant cluster (a momentum / macro-rank mixture, MDA +0.009) and a top intraday cluster (`f10_oc_ret`, `f7_overnight_gap`, MDA +0.009). The Bloomberg ablation (`bbg_missingness_ablation.csv`) now shows a near-flat BBG contribution at the per-class level ($\text{delta} \leq 0.001$ on every class) once features fall to 67–68 columns after the drift filter and the pooled random-forest is selected as best model; the with-BBG configuration is shipped on parsimony. Three instruments (`ho1s`, `ng1s`, `rb1s`) carry explicit coverage caveats; three (`hg1s`, `rb1s`, `si1s`) have the SOPS gate land at $p^* \geq 0.5$ and therefore contribute no test-period positions.

The strategy verdict is consistent across five robustness lenses. AUC at the per-class level beats random on equity and energy; cluster importance shows MDA-significant clusters on equity and metals (energy dominated by a single significant cluster); bootstrap Sharpe excludes zero; the DSR ladder stays above 0.94 even at $4 \times$ the raw trial count; and Pesaran–Timmermann attains $p < 10^{-5}$. Reported gross numbers should be read alongside the transparent cost block — 304 bps of total transaction cost is already deducted from the headline 16.2 % annualised net return.

Figure 1. Pipeline architecture and report-section map. Sage panels denote OHLCV-only families; warm panels denote Bloomberg-derived families. Stage numbers in the left column correspond to report sections.



1. Feature Engineering [20 marks]

The feature panel is the union of 16 self-registered families that collectively register 94 columns; the drift filter ($KS \leq 0.25$ or $\text{train} \rightarrow \text{val AUC} \geq 0.54$) admits 67–68 columns per asset class into the modelling matrix. Every feature is a small pure function $f(\text{panel}, \text{ctx}) \rightarrow \text{Series}$ that takes a frozen `FeatureContext` (the per-instrument frame, the cleaned Bloomberg parquets, and the asset-class membership map) and self-registers a `FeatureSpec` at module load via `register` in `features/catalog.py`. The registry is frozen at import time; adding a feature requires a new module-level `register(...)` call. The assembler iterates the registry once per instrument, reindexes each feature onto the instrument’s trading-day index, and emits one row per (instrument, date) on the intersected calendar.

Each `FeatureSpec` carries a leakage class. *E-class* features are *causal by truncation invariance* — their value at bar t depends only on the panel up to bar t , verified by the truncation test in `tests/experimental/test_features.py`. *TF-class* features (only `f17_hmm_state_lo/_mid/_hi`) fit on the modelling-train block ($\text{date} \leq 2021-07-01$) and are applied with frozen parameters; that fit window cannot slide into validation or test. This contract is enforced at the assembler level rather than relying on discipline at use sites.

1.1 The 16 feature families

Table 1 groups the 16 families along the eight economic axes that span the rubric. Counts in parentheses are the registry sizes read from `features/catalog.py`; the total of 94 columns is the registered count, of which 64 survive the global drift filter and 67–68 enter each per-class modelling matrix after class-specific identity additions. Two prototyped families — **F18 term structure** (5 features) and **F19 options IV** (6 features) — were dropped from the registry on parsimony after the cluster importance analysis (`features/catalog.py` carries them as commented-out `register` lines with the rationale).

1.2 Bloomberg integration: forensic data audit before use

The Bloomberg block was added *after* a forensic audit (`reports/submission/bloomberg_validation_report.md`) discovered two issues that would have produced silently wrong features had they not been caught.

Issue 1: OHLCV is back-adjusted, BBG is raw. The competition OHLCV panel applies continuous-futures roll adjustment (CL closes at 24.80 on 2020-01-02 against a raw BBG `LAST_PRICE` of 61.18). Computing F18 as $(\text{BBG}_{2nd} - \text{OHLCV}_{front}) / \text{OHLCV}_{front}$ would produce term-structure spreads on the order of $\pm 100\%$. The fix: F18 is computed entirely from the BBG front + BBG second parquet, never mixed with OHLCV. F1–F17 + F21 remain on OHLCV (the investable, continuous-return series), and F18–F19 + F22 remain on raw BBG.

Issue 2: EIA crude inventory is dated by as-of, not release. The `DOEASCRD` ticker carries the Friday as-of date of the EIA Weekly Petroleum Status Report; the figure is released the *following Wednesday* at 10:30 AM ET. A naive PIT alignment would leak roughly five calendar days of forward information. The cleaner applies an explicit `EIA_LAG_DAYS = 5` shift to both `f22_eia_crude_change_z26w` and the `f22_eia_release_flag` binary, aligning the feature to the release-day calendar rather than the measurement-week calendar.

Substitutions documented in the methodology. (i) The BBG IV pull uses 90%/110% moneyness IV rather than 25 Δ put/call IV — captures the same skew direction within $\sim 10\text{--}20\%$ relative error; (ii) Platinum (XPT) options data is missing in the pull, so `p11s.f19_*` falls back to GC1 IV (XPT options market is thin and this fallback was pre-registered); (iii) the CFTC COT family (F20) is dropped entirely because it was not in the pull; (iv) F22 is partial (EIA only — no FOMC, OPEC, USDA, ECB). These decisions are recorded in the report and propagated to `outputs/coverage_caveat.csv`.

Table 1. Feature manifest — 16 families registered in `features/catalog.py`, with their counts and leakage class. **E** = causal by truncation invariance; **TF** = train-fitted, frozen at inference; **B** = Bloomberg-derived. **94 columns** total register; 64 survive the global drift filter and 67–68 enter the per-class modelling matrix.

Family	Captures	Cnt	Cls	Representative members
F1 technical	mean-reversion oscillators	3	E	rsi_14, bb_pctb_20
F2 volatility	realised / Parkinson / GK / vov	6	E	vol_20/60, vol_of_vol_20
F5 signal	primary-signal dynamics	6	E	trailing_run_length, flip_rate_60
F6 momentum	TS momentum + MACD	6	E	ts_momentum_20/60, macd_12_26
F7 liquidity	volume / OI / Amihud / Kyle	7	E	oi_level, volume_z_20, amihud_20
F8 calendar	day-of-week, month	4	E	dow_sin/cos, month_sin/cos
F9 cross-section	dispersion / pair-corr	4	E	xsect_rank_60, pair_corr_63
F10 intraday	HL range, open-close	4	E	hl_range, oc_ret
F11 macro B	VIX, MOVE, DXY, rates, OAS	35	E	vix_rank63, ust10_chg5, hy_oas_chg20
F12 path	efficiency / var-ratio / Hurst	5	E	efficiency_ratio_21, hurst_100
F15 risk	semi-vol / tortuosity	2	E	semi_vol_ratio_20
F16 regime alignment	regime score	1	E	regime_alignment_score
F17 HMM (regime)	3-state HMM filtered posteriors	3	TF	hmm_state_lo/_mid/_hi
F21 cross-asset	gold/silver / copper/gold / crack	4	E	gold_silver_z63, crack_321
F22 events B	EIA crude flag + change	2	E	eia_release_flag, eia_crude_change
EWMA-HMM	high-vol prob + switch prob	2	E	prob_highvol, switch_prob
Total registered		94		
Kept post drift filter		64		(67–68 per class with class-specific additions)

1.3 Macro features as *rolling ranks*, not levels

A pre-modelling diagnostic on macro *levels* (VIX, MOVE, DXY, 10Y UST, BE10Y) showed median KS statistic of 0.358 between the modelling train and test windows (max 1.000 for the level series). Levels with $KS \rightarrow 1$ are useless for cross-window inference: the same VIX of 22 encodes different regime information in 2020 versus 2022. The macro family in `features/macro.py` therefore reformulates every level as a 63-day rolling rank in $[0, 1]$ via `rolling_rank` (e.g. `f11_vix_rank63`, `f11_dxy_rank63`). *Changes* (5-day and 20-day differences such as `f11_ust10_chg5`, `f11_be10y_chg20`) are left as-is because the differencing operator already removes the dominant non-stationarity. This is consistent with the concept-drift framing of Gama et al. (2014): the $P(X)$ shift in macro indicators is partly absorbed by the per-window rank transformation.

1.4 Drift filter and KS audit

The drift filter in `make_features.py` computes the two-sample KS statistic between an early-train (first 70 % chronologically) and late-train (last 30 %) slice for each feature, plus a direction-agnostic single-feature val-AUC on the late-train slice. A feature is dropped if $KS > 0.25$ and $val\text{-}AUC < 0.54$. The drop rule is intentionally permissive: validation and test partitions are sealed during this step so they remain honest evaluators downstream. Drift-filtered features include `f19_iv_pctile_252` (KS 0.626, val-AUC 0.532), `f2_vol_20`, `f2_parkinson_20`, and `f2_garman_klass_20`; each is kept-but-flagged or dropped per the rule rather than silently zeroed. The full ledger is `results/submission/feature_drift_audit.csv`.

1.5 Cross-asset and signal-dynamics features

Five F21 features capture systemic information invisible to per-instrument engineering: gold/silver ratio z-score (`f21_gold_silver_z63`), copper/gold ratio (`f21_copper_gold_ratio`), and the 3-2-1 crack spread (`f21_crack_321`); asset-class membership is encoded separately under identity one-hots elsewhere. Six F5 signal-dynamics features condition the metamodel on the structure of the primary signal itself — not on its sign, but on its persistence: `trailing_run_length`, `days_since_flip`, `flip_rate_60`, `signal_entropy_20`, `long_bias_20`, `participation_60`. Together they let the model condition on *whether a fresh signal in a stable regime* looks different from *a signal during oscillation*, without re-using the sign at the prediction step (`f5_long_bias_20` is a population statistic, not the current bet).

1.6 Feature panel summary

After the drift filter, 67–68 columns enter the per-class modelling matrix under *with-BBG* and 65–66 under *without-BBG*; the small gap reflects that several BBG features are already drift-filtered out before selection. The Bloomberg ablation (Table 2) reads delta close to zero on every asset class once the random-forest pool is selected as best model: the BBG features are not subtractive on any class but they are not pivotal either. The with-BBG configuration is shipped on the absence of a negative delta, with the parsimonious caveat that the ablation does not show evidence of a positive BBG contribution that would not also be captured by the engineered F1–F17 panel.

Table 2. Bloomberg ablation — per-class random-forest CPCV(6,2) AUC with and without Bloomberg-derived features, plus a simulated-missingness control that re-runs with-BBG features but zeros out BBG columns at inference time. The with-BBG configuration is shipped on the delta-close-to-zero evidence. Source: `bbg_missingness_ablation.csv`.

	with-BBG	w/o BBG	sim. miss.	Δ wo – with	Ship
Equity	0.550	0.550	0.550	0.000 0	with-BBG
Energy	0.670	0.670	0.671	–0.000 3	with-BBG
Metals	0.538	0.538	0.538	0.000 0	with-BBG

2. Triple-Barrier Labelling [20 marks]

The shipped labels are *per-instrument* triple-barrier labels selected from a 343-cell grid $(p_t, s_l) \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 2.5\}^2$ and $h \in \{1, 2, 3, 5, 10, 15, 20\}$ by an adjusted-Sharpe grid search on the *development partition only* — the test partition is never touched during label selection. The label is binary $y_{i,t} \in \{0, 1\}$ following López de Prado (2018) Ch. 3, taking value 1 if the side-adjusted price path hits the profit-take barrier before the stop-loss within the vertical window of h trading days. The labelling logic itself is identical to the canonical AFML implementation; the per-instrument grid search replaces the global symmetric default with geometries adapted to each instrument’s return distribution.

2.1 Per-instrument geometries adopted

Table 3. Per-instrument barrier geometries adopted, with the realised touch profile on the development partition. Source: `per_instrument_geometry_summary.csv`. **Bold** geometries are the five instruments with $h \geq 10$ that the EDA argues are not self-fulfilling.

Instrument	p_t	s_l	h	n_{events}	pos. rate	frac PT	frac SL	frac vert.
cl1s	0.25	0.25	1	421	0.599	0.480	0.297	0.223
es1s	1.00	0.25	10	565	0.427	0.427	0.573	0.000
fesx1s	0.25	0.25	1	636	0.535	0.425	0.358	0.217
gc1s	1.00	0.25	15	160	0.500	0.500	0.500	0.000
hg1s	1.00	0.25	10	618	0.413	0.409	0.587	0.003
ho1s	0.25	0.25	1	63	0.667	0.540	0.270	0.190
ng1s	0.25	0.25	1	124	0.524	0.411	0.395	0.194
nq1s	0.25	0.25	1	603	0.539	0.448	0.357	0.196
pl1s	0.25	0.25	1	556	0.540	0.466	0.362	0.173
rb1s	2.50	0.25	15	613	0.263	0.251	0.737	0.011
si1s	0.75	0.25	20	558	0.414	0.414	0.586	0.000
Total				4,917				

Three structural observations from the geometry grid carry directly into the rest of the pipeline:

(i) $s_l = 0.25$ is *universal*. The tight stop-loss dominates every winner. This is consistent with the futures return distribution heavy left tail: a wide stop quickly accumulates rare large losses that swamp the positive-skew profits of a well-chosen profit-take. The label as optimised is therefore a *positive-skew profile* with frequent small profits, frequent small stops, and rare vertical exits — exactly the trade-distribution that a meta-model can filter.

(ii) Six of eleven instruments pick $h = 1$ (*cl1s*, *fesx1s*, *ho1s*, *ng1s*, *nq1s*, *pl1s*). At $h = 1$ the label is largely a function of the first forward bar, which is the same bar as the lag-1 return. Filtering lag-1 PnL by y is therefore *partly circular* — the EDA in `triple-barrier-label.pdf` flags these as “self-fulfilling”. The five $h \geq 10$ instruments (**es1s**, **gc1s**, **hg1s**, **rb1s**, **si1s**) are trusted most for forward-looking claims.

(iii) **rb1s** positive rate is 0.26 versus the typical 0.5. The $p_t = 2.5$ choice makes the profit barrier ten times wider than the stop, so the label is heavily skewed toward SL hits. The model must explicitly handle this imbalance — the `balanced_sample_weight` convention in `models.py` rescales each class to carry mass n/n_{classes} while preserving within-class uniqueness proportions.

2.2 Sigma source, entry convention, and barrier widths

The barrier widths at event t are $p_t \cdot \hat{\sigma}_{i,t} \cdot \sqrt{h}$ (PT) and $-s_l \cdot \hat{\sigma}_{i,t} \cdot \sqrt{h}$ (SL), in signed-bet log-return units. The volatility estimator $\hat{\sigma}_{i,t}$ is `f2_vol_20` — a 20-day rolling close-to-close std — consistent

across all 11 instruments per `label_outcome_audit.csv` (`sigma_source` column reports `f2_vol_20` uniformly). The entry convention used by the shipped labels is $t_{\text{start}} = t_{\text{signal}}$: the signal observed at the close of bar t is acted on at the close of bar t itself, and the realised side-adjusted return is $\text{sgn}(s_{i,t}) \cdot (p_{t_{\text{end}}}/p_{t_{\text{start}}} - 1)$. This is verified empirically against the `ret` column in the labels CSV. The next-bar empirical regularity that motivates a $t + 1$ interpretation is captured by the small h choices in the grid, not by shifting the entry index.

2.3 Sample uniqueness

Per-event sample uniqueness is computed per López de Prado (2018) Ch. 4 from the indicator matrix of which trading bars each labelling window touches. Six instruments are at $h = 1$ and therefore have *mean uniqueness* = 1.000 (consecutive events do not overlap, so each event is fully unique). The larger- h instruments overlap and have mean uniqueness materially below 1: `es1s` 0.626, `gc1s` 0.629, `hg1s` 0.590, `rb1s` 0.370, `si1s` 0.650. `rb1s` carries the lowest uniqueness because $h = 15$ vertical windows overlap heavily; this is why `balanced_sample_weight` multiplies the uniqueness against the inverse class frequency before being passed as `sample_weight` to the underlying classifiers — preserving both the overlap correction and the `rb1s` class imbalance correction simultaneously.

2.4 Partition discipline

The labels CSV ships an authoritative `partition` column with values `train` (~60%), `val` (~21%) and `test` (~19%) determined chronologically per instrument. The pipeline honours this partition everywhere: champion selection runs CPCV(6,2) on `train` with `val` as honest scoreboard; calibration fits on the OOF `train` + `val` predictions; the H1 2022 deliverable is exactly the `test` partition. `train` + `val` are combined for the final deliverable refit so the predict-time model uses ~24 months of labels rather than ~18. The `embargo_p90` field in `instrument_scope.json` extends each instrument’s purged-CV embargo to the 90th percentile of its own label-span distribution, eliminating the near-edge leakage that a global 1% embargo would miss for the longer- h instruments.

2.5 Oracle headroom: what a perfect meta-classifier could lift

A theoretical-ceiling diagnostic motivates whether a meta-model is worth building at all on this label set. Per instrument, the cumulative side-adjusted log-return summed over all H1 2022 OOS events is compared under two scenarios: *primary signal* (take every signal day) and *oracle* (take only events where the realised label is 1). The gap is the per-event return space available to any (imperfect) meta-classifier filter.

The pooled oracle gap of +8.76 is the upper bound on what *any* meta-classifier filter on this event set could lift cumulative event return by. The realised SOPS deliverable captures a small fraction of this gap (consistent with the test AUC of 0.49–0.67 across asset classes) but the gap’s existence — and its persistence into the sealed test slice — is what justifies the meta-modelling exercise. The two negative-primary instruments (`ho1s`, `si1s`) are especially illustrative: a perfect filter takes the primary signal from loss-making to break-even or positive without otherwise changing the trade list.

Table 4. Oracle-vs-primary cumulative log-return per instrument over the H1 2022 OOS window, summed across all events. The **gap** column is the per-event return space available to a meta-classifier. Source: `oos_events_with_predictions.csv` aggregated by instrument.

Instrument	Primary	Oracle	Gap
clls	0.548	1.211	0.663
esls	0.110	1.021	0.911
fesxls	0.340	0.965	0.625
gcls	0.134	0.208	0.074
hgls	0.316	1.189	0.873
hols	−0.070	0.000	0.070
ngls	0.204	1.308	1.104
nqls	0.410	1.306	0.896
plls	0.154	0.995	0.841
rbls	0.972	2.630	1.658
sils	−0.072	0.975	1.047
Pooled	3.046	11.807	+8.760

3. Model Development & Comparison [30 marks]

The same modelling matrix — 67–68 columns per asset class after the drift filter — is fed through a four-candidate model roster crossed with three candidate pool structures per instrument. The champion is selected by mean CPCV(6,2) AUC with a 1-SE rule (most regularised within one standard error of the best).

3.1 Model families

Five estimators behind one uniform `MetaClassifier` interface in `models.py`:

- **elastic-net logistic** (`make_elasticnet_logistic`) — standardised inputs, median-imputed NaNs, fixed `l1_ratio = 0.5` and `C = 1.0` (the default mix, not tuned inside the CPCV inner loop). The linear baseline; cheap to fit and interprets the feature panel coefficient-wise.
- **XGBoost** (`make_xgb`) — the binary-logistic objective with `max_depth=4`, `lr=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, $\alpha = 0.1$, $\lambda = 1.0$, single-threaded for byte-stable re-fit. **Consumes NaNs natively** — critical for the BBG-missingness ablation, where the simulated-missingness arm zeros BBG columns at inference time and XGBoost handles them without imputation.
- **LightGBM** — gradient-boosted decision trees with leaf-wise growth; the tree-based competitor to XGBoost.
- **random forest** (`make_random_forest`) — `RandomForestClassifier` with `max_features='sqrt'` (the explicit choice rather than 'auto' for sklearn ≥ 1.3 compatibility).
- **multi-task NN** (`multitask.MultiTaskMetaClassifier`) — a shared encoder with per-instrument heads inside a sub-class pool, trained with sample-weighted binary cross-entropy.

Plus an `ensemble_simple` equal-weight average that is also evaluated inside the four-candidate roster.

3.2 Pool structure

Each instrument has between one and three candidate pools (defined in `INSTRUMENT_REGIMES` in `champion_pipeline.py`):

- *Equity* (`es1s`, `nq1s`, `fesx1s`): individual versus `equity_all`. Already specialised within the class; pooling helps only when within-class commonality dominates within-instrument idiosyncrasy.
- *Energy*: `cl1s` chooses among individual, `energy_cl_ho` (CL+HO), and `energy_all`; `ho1s` between `energy_cl_ho` and `energy_all` (too few events for individual); `rb1s` between individual and `energy_all`; `ng1s` only `energy_all`.
- *Metals*: `gc1s`, `si1s`, `pl1s` each among individual, `precious` (GC+SI+PL), and `metals_all`; `hg1s` between individual and `metals_all` (HG sits structurally outside the precious trio).

This per-instrument pool structure is the meta-modelling counterpart of forecast combination under regime heterogeneity (Claeskens et al. 2016): a pooled fit reduces variance from small-sample noise but biases the model toward the cross-instrument average. The 1-SE rule formalises this trade-off by preferring the most regularised candidate that is within one standard error of the best mean AUC. The CV scheme is the `CombinatorialPurgedCV` class implementing the López de Prado (2018) Ch. 12 design with $N = 6$ groups and $k = 2$ test groups per fold, giving $\binom{6}{2} = 15$ purged paths per candidate.

3.3 Champion AUC by instrument

Seven of eleven instruments clear AUC 0.55 mean. Six of eleven clear the 1-SE lower bound at 0.50, with `ng1s` the strongest (**0.619 [0.569, ...]**). The two below-0.55 instruments are `fesx1s` (0.516, equity-index correlated to `es1s`) and `si1s` (0.512, weakest within the precious

Table 5. Per-instrument champion (with-BBG configuration). **Pool**, **Model**, **Champion AUC** (CPCV(6,2) 15-path mean), **SEM** across paths, **1-SE lower CI**, and **Selection reason**. Source: `champions_summary.csv`.

Instr.	Pool	Model	AUC	SEM	1-SE lo	Reason
es1s	equity_all	elastic-net	0.548	0.013	0.535	1-SE pick
nq1s	nq1s	random_forest	0.557	0.012	0.544	1-SE pick
fesx1s	equity_all	elastic-net	0.516	0.009	0.507	1-SE pick
cl1s	energy_all	multi-task NN	0.566	0.015	0.551	best mean
ho1s	energy_cl_ho	LightGBM	0.552	0.054	0.498	best mean
rb1s	rb1s	random_forest	0.526	0.013	0.512	1-SE pick
ng1s	energy_all	elastic-net	0.619	0.050	0.569	best mean
gc1s	metals_all	LightGBM	0.566	0.020	0.547	best mean
si1s	metals_all	multi-task NN	0.512	0.012	0.500	best mean
pl1s	pl1s	LightGBM	0.564	0.014	0.550	best mean
hg1s	metals_all	elastic-net	0.539	0.017	0.522	best mean

trio). `ho1s` clears in mean (0.552) but *not* in 1-SE lower bound (0.498) because the test partition for `ho1s` has only 2 events (one of the explicit caveats in `coverage_caveat.csv`, propagated to a `thin_oos` flag and excluded from any quantitative test-set claim).

3.4 Per-class XGBoost benchmark

Table 6. Per-class baseline (pooled CPCV(6,2) AUC). The four-candidate roster ran XGBoost, LightGBM, random forest, and elastic-net; the **best model** column reports the winner on the modelling sample by mean OOS AUC. Random forest is the per-class winner across all three asset classes and across all three ablation arms. Source: `baseline_xgb_per_class.csv`.

Class	Variant	Best model	$n_{\text{modelling}}$	n_{feat}	Pooled AUC	std
equity	with-BBG	random forest	1 083	67	0.550	0.020
equity	without-BBG	random forest	1 083	65	0.550	0.020
equity	sim. missingness	random forest	1 083	67	0.550	0.020
energy	with-BBG	random forest	668	68	0.670	0.034
energy	without-BBG	random forest	668	66	0.670	0.043
energy	sim. missingness	random forest	668	68	0.671	0.037
metals	with-BBG	random forest	1 174	68	0.538	0.043
metals	without-BBG	random forest	1 174	66	0.538	0.043
metals	sim. missingness	random forest	1 174	68	0.538	0.043

3.5 Why random forest dominates the per-class benchmark

The per-class benchmark (Table 6) is settled by the random-forest pool on every asset class. Two reasons make this unsurprising on the modelling sample. First, with $n_{\text{modelling}}$ between 668 and 1,174 and 67–68 features after the drift filter, the ratio is borderline for gradient-boosted ensembles (XGBoost / LightGBM) whose leaf-wise growth needs more rows to localise non-linear interactions; random forest’s bagged-tree averaging is more sample-efficient at this scale. Second, the per-instrument geometries (`rb1s` at $p_t = 2.5$, six instruments at $h = 1$) create label distributions with heavy class imbalance and short signal horizons; random forest’s coarse interaction recovery handles these regimes more robustly than the boosted competitors.

The per-instrument champion table (Table 5) still selects a mix of elastic-net, LightGBM, random forest, and multi-task NN because the 1-SE rule is applied *within instrument* rather than at the pooled level. The per-instrument champions are calibrated with Platt scaling on their own purged OOF predictions (Section 5) and aggregated into one OOF frame for the strategy layer.

3.6 Multi-task NN: hyperparameter tuning grid

The brief calls explicitly for a neural-network family with hyperparameter tuning. The multi-task NN candidate (`multitask.MultiTaskMetaClassifier`) is a shared encoder with per-instrument heads trained with sample-weighted binary cross-entropy. A four-configuration grid sweeps capacity, regularisation, and learning rate around the locked production point. The grid is scored under CPCV(6, 2) on the largest cross-instrument pool (`equity_all`, 3 instruments, ~1,800 events) by `scripts/tune_multitask_nn.py`; results are persisted to `results/submission/multitask_nn_tuning.csv`.

Table 7. Multi-task NN hyperparameter sweep on `equity_all` pool (CPCV(6,2), 14 paths). The locked production configuration sits within the 1-SE band of the in-grid best. Source: `multitask_nn_tuning.csv`.

Config	hidden	drop	lr	epochs	mean AUC	SEM	1-SE of best
locked (production)	[64,32]	0.10	0.000 5	100	0.531 2	0.007 7	✓
wider capacity	[128,64]	0.10	0.000 5	100	0.517 3	0.007 7	
more dropout	[64,32]	0.20	0.000 5	100	0.534 4	0.008 2	✓
higher lr	[64,32]	0.10	0.001 0	100	0.532 0	0.007 4	✓

The in-grid best is `more_dropout` at 0.5344 mean AUC; the locked configuration (0.5312) sits within one standard error of it. Wider capacity (0.5173) is the only configuration that falls outside the 1-SE band, consistent with the small-sample over-parameterisation hypothesis from §3.3. The production configuration is retained on the 1-SE rule rather than swapping to the marginally-higher `more_dropout` variant.

3.7 Deployment substitution: NN champions on `c11s` / `si1s`

The CPCV champion picker (`champion_pipeline.py`) selects the multi-task NN for `c11s` (`energy_all` pool, AUC 0.566) and `si1s` (`metals_all` pool, AUC 0.512). The shipped deliverable substitutes `elasticnet_logistic` on these two instruments inside `make_deliverables.py` so the prediction pipeline does not require PyTorch at inference time. The substitution is the 1-SE rule’s second choice for those instruments; the AUC delta on each is within the champion-table SEM. The CPCV champion identification, however, retains the multi-task NN on the modelling sample — which is what Table 5 reports and which evidences the NN family’s participation in the four-candidate roster required by the brief.

4. Cluster-Level Feature Importance [10 marks]

Feature importance is reported *at the cluster level* rather than per-feature, because the panel contains many correlated features within each economic family. A per-feature permutation MDA is then dominated by within-family correlation: permuting `f2_vol_20` while leaving `f2_vol_60` untouched understates the joint contribution of the volatility family. Cluster MDA permutes all members of an average-linkage cluster jointly and reports the AUC drop on the held-out validation slice; the rank is then cross-checked against MDI (XGBoost gain) and SHAP (TreeSHAP magnitude). Reported numbers come from `importance/{equity,energy,metals}/cluster_crosscheck_table.csv`.

4.1 Cluster definition and rank agreement

For each asset class, the training-set feature correlation matrix is clustered using average-linkage on $d_{ij} = \sqrt{0.5(1 - \rho_{ij})}$ at $K = 16$. Three importance metrics are computed and ranked: cluster MDA mean across 500 permutation repeats, cluster MDI sum (XGBoost gain summed across members), and cluster SHAP sum (TreeSHAP magnitude summed across members). Rank agreement is quantified by Kendall- τ in `rank_agreement.csv` per class. Significance is flagged at MDA mean $> 1\sigma$ across permutation repeats.

4.2 Per-class top clusters

Table 8. Top-3 clusters by MDA mean per asset class, with member counts and the three importance metrics. Significance flag is MDA $> 1\sigma$ across permutation repeats. Source: `importance/<class>/cluster_crosscheck_table.csv`.

Class	C#	Mem	MDA	MDI	SHAP	Sig	Representative members
equity	12	6	0.021	0.135	0.346		rsi_14, bb_pctb_20, ts_mom_20, macd, semi_vol_ratio
equity	16	7	0.009	0.120	0.318	✓	ts_mom_60, ma_cross, macd, skew_rank63, ig_oas_rank63, lme_copper_rank63
equity	10	4	0.009	0.070	0.274	✓	overnight_gap, dow_sin/cos, oc_ret
energy	7	3	+0.129	0.189	0.837	✓	participation_60, oi_level, ewma_hmm_prob_highvol
energy	1	6	0.007	0.091	0.357		bb_pctb_20, rsi_14, ts_mom_20, trend_tval_21
energy	6	3	0.003	0.044	0.170		be10y_chg20, ust10_chg5, be10y_chg5
metals	2	6	0.007	0.106	0.257		long_bias_20, ts_mom_60, ma_cross, macd, xsect_rank_60
metals	1	7	0.007	0.145	0.370		bb_pctb_20, rsi_14, ts_mom_20, macd_hist, trend_tval_21
metals	7	2	0.004	0.015	0.101	✓	dow_sin, dow_cos

Three readings follow from the cluster picture.

(i) *Energy is dominated by a single significant cluster.* Cluster 7 in energy — `f5_participation_60` + `f7_oi_level` + `ewma_hmm_prob_highvol` — has cluster MDA **0.129**, twenty times the next-strongest energy cluster and an order of magnitude above any significant cluster on equity or metals. The mean SHAP-magnitude on `f7_oi_level` alone is 0.602 and on `f5_participation_60` is 0.523; both substantially above the panel-average. Energy meta-labelling thus rests on a microstructure-plus-regime signal (open interest, signal participation rate, high-vol HMM posterior) rather than on price-based momentum or mean-reversion — and this finding explains why the per-class energy AUC sits at 0.670 versus 0.538–0.550 for the other classes.

(ii) *Equity admits two significant clusters — both momentum and intraday.* Cluster 16 in equity — a

momentum-and-macro-rank mixture (f6_ts_momentum_60, f6_ma_cross_20_60, f6_macd_12_26, f11_skew_rank63, f11_lme_copper_rank63) — registers MDA +0.009 at $> 1\sigma$; cluster 10, the intraday-microstructure block (f7_overnight_gap, f10_oc_ret, f8_dow_sin/cos), also significant at +0.009. The largest cluster MDA on equity — cluster 12 at +0.021 (the mean-reversion oscillator family) — is *not* flagged significant, indicating high within-cluster permutation variance. The pruned-vs-full ablation gives **full AUC 0.531 vs pruned AUC 0.552** on a 6-feature pruned set — a 2.1pp lift on equity from removing 51 columns, less aggressive than the prior measurement window’s 6.9pp result but still favouring a sparser representation.

(iii) *Metals admits a single significant cluster.* Cluster 7 in metals — f8_dow_sin, f8_dow_cos (calendar effects) — is the only flagged cluster at MDA +0.004 ($> 1\sigma$). The two largest clusters by MDA mean (mean-reversion C1 at +0.007 and a momentum/cross-sectional C2 at +0.007) are not significant. The full-vs-pruned ablation on metals is negative (0.553 vs 0.526, $\Delta = -0.027$), suggesting the metals panel is *not* amenable to aggressive sparsification under the current pruner.

4.3 Pruned-vs-full ablation

Table 9. Pruned-vs-full validation AUC. Pruned set selected by taking the cluster’s top-PC representative for any cluster whose PC1 variance is $> 50\%$ plus the global SHAP top-5. Source:

importance/<class>/pruned_vs_full_auc.csv.

Class	Full				Pruned				Δ AUC
	n_f	n_v	AUC	SEM	n_f	n_v	AUC	SEM	
equity	64	364	0.531	0.030	6	364	0.552	0.030	0.021
energy	64	297	0.611	0.033	5	297	0.603	0.033	−0.008
metals	64	380	0.553	0.030	5	380	0.526	0.030	−0.027

The equity pruned-vs-full lift of +2.1pp is modest but still positive; the energy ablation is essentially flat (−0.008, well within one SEM); the metals ablation is meaningfully negative at −0.027, meaning that on metals the 5-feature pruned set under-performs the 64-feature full panel and the panel cannot be aggressively sparsified without losing signal. *The pruned set is not the shipped model* — it is a diagnostic that quantifies the over-fitting risk implicit in the full panel and motivates the elastic-net candidate in the model roster as a mid-point between full and pruned. The shipped per-instrument champions are selected under the 1-SE rule precisely to bias toward this regularisation effect without committing to a single hand-pruned set.

5. Model Evaluation [20 marks]

Five evaluation channels report on the deliverable: per-instrument classification metrics on the modelling sample, per-class benchmarks, threshold analysis under the bootstrap- p^* gate, headline strategy significance on the sealed H1 2022 test partition with stationary block-bootstrap and Lo-Opdyke analytic CI, and a deflation ladder (PSR, MinTRL, DSR) sized to the actual hyperparameter search trial count.

5.1 Per-instrument classification metrics

Table 10. Per-instrument modelling-sample classification metrics under the winning per-class model (random forest across all three classes — Table 6). Source: `baseline_per_instrument.csv`.

Instr.	Class	n	pos. rate	AUC	log-loss	Brier	Acc	Prec	Rec	F1
es1s	equity	1 612	0.452	0.547	0.696	0.251	0.538	0.490	0.543	0.515
fesx1s	equity	1 779	0.531	0.522	0.701	0.253	0.527	0.551	0.589	0.570
nq1s	equity	1 661	0.536	0.566	0.686	0.246	0.550	0.579	0.588	0.583
cl1s	energy	1 066	0.601	0.495	0.691	0.248	0.579	0.598	0.911	0.722
ho1s	energy	160	0.650	0.457	0.710	0.258	0.544	0.650	0.644	0.647
ng1s	energy	140	0.500	0.532	0.718	0.261	0.521	0.511	1.000	0.676
rb1s	energy	1 750	0.189	0.520	0.542	0.178	0.789	0.232	0.051	0.083
gc1s	metals	566	0.472	0.503	0.709	0.258	0.483	0.469	0.709	0.564
hg1s	metals	1 753	0.391	0.487	0.694	0.250	0.530	0.370	0.285	0.322
pl1s	metals	1 576	0.557	0.504	0.692	0.249	0.530	0.551	0.849	0.668
si1s	metals	1 581	0.420	0.478	0.704	0.255	0.481	0.397	0.448	0.421

The per-instrument distribution makes the pooled per-class AUC tell a finer story. **ho1s** (AUC 0.457), **si1s** (0.478), **hg1s** (0.487), and **cl1s** (0.495) sit *below* random on their per-instrument modelling samples; **ho1s** is explicitly caveated (160 events, thin) and is excluded from any quantitative test-period claim. The pooled energy benchmark of 0.670 (Table 6) is *higher* than every individual energy instrument because the pooled fit exploits cross-instrument structure (the `f5_participation_60` / `f7_oi_level` cluster, Section 4) that does not show up cleanly on any single instrument’s random-forest probit calibration. The strongest per-instrument classifications are **nq1s** (AUC 0.566) and **es1s** (0.547); **ng1s** (0.532, $n = 140$) is the only energy instrument above 0.50 on the individual sample.

5.2 Per-instrument Platt calibration and bootstrap p^* gate

After per-instrument champion selection (Section 3), a Platt sigmoid (LogReg-based) is fit *per-instrument* to the OOF predictions on `train + val` (`calibration.PlattCalibrator`) and applied at predict time on the sealed test. Platt is monotone $\Rightarrow$ AUC is unchanged before/after calibration (verified by unit test); only Brier, log-loss, ECE and the sizing stake are affected. The bootstrap p^* gate then sets the meta-decision threshold per instrument as $p^* = L/(G + L)$ where G is the mean TP return and L the mean FP loss across bootstrap resamples of the OOF (slide 21).

Six of eleven instruments actively contribute to the test window. **es1s** clears the gate threshold but the calibrated probabilities are below it for every test event ($p^* = 0.385$ is unusually low, reflecting a small G/L asymmetry from the wide $p_t = 1.0$ profile). **hg1s**, **rb1s**, and **si1s** have $p^* = 0.500$ — the bootstrap collapsed to a degenerate threshold because the OOF distribution did not contain enough TP-FP separation to compute $p^* = L/(G + L)$ stably; the gate then routes weight to zero. The strategy is therefore driven by **cl1s**, **fesx1s**, **gc1s**, **ng1s**, **nq1s**, and **pl1s** on the test window.

Table 11. Per-instrument bootstrap p^* gate and the resulting test-window position activity under the SOPS sizer. Source: `threshold_summary.csv`.

Instr.	p^*	n_{oos}	n_{taken}	mean $ w $	max $ w $	Status
cl1s	0.453	88	88	0.135	0.295	all-taken
es1s	0.385	109	0	0.000	0.000	p^* never crossed
fesx1s	0.514	126	124	0.420	0.604	near-all-taken
gc1s	0.281	22	15	0.451	0.800	68 % taken
hg1s	0.500	115	0	0.000	0.000	no positions (degenerate p^*)
ho1s	0.549	2	2	0.172	0.172	thin OOS (excluded)
ng1s	0.423	56	45	0.099	0.145	80 % taken
nq1s	0.457	122	122	0.347	0.526	all-taken
pl1s	0.507	104	68	0.197	0.353	65 % taken
rb1s	0.500	110	0	0.000	0.000	no positions (degenerate p^*)
si1s	0.500	97	0	0.000	0.000	no positions (degenerate p^*)

Table 12. Significance battery on the SOPS deliverable, H1 2022 sealed test window. All metrics on per-period net returns. Source: `significance_summary.csv`.

Statistic	Value	Reading
n (periods)	129	sealed H1 2022
Per-period Sharpe SR	0.198	uncorrected for n
Annualised Sharpe ($\times \sqrt{252}$)	3.147	valid if Ljung-Box clears
$t = \text{SR} \cdot \sqrt{n}$	2.251	> 1.96 , significant at 5 %
Bootstrap 95 % CI per period	[0.022, 0.356]	excludes zero
Bootstrap block length (Politis–White)	4.47	data-driven
Lo/Opdyke analytic 95 % CI per period	[0.028, 0.369]	parametric cross-check
PSR($\text{SR}^* = 0$)	0.989	> 0.95 deploy threshold
MinTRL (95 % PSR)	68	$< n$, certified
Ljung-Box $Q(10)$	13.27 ($p = 0.21$)	IID-like, $\sqrt{252}$ valid
Pesaran–Timmermann	$S = 4.60$ ($p < 10^{-5}$)	positive directional skill
Treynor–Mazuy γ	-0.253 ($t = -4.05$)	convex timing (read with scale caveat)
Henriksson–Merton hit	0.571 ($z = 4.38$)	base-rate sensitive

5.3 Headline strategy significance

The bootstrap 95 % CI on per-period Sharpe excludes zero. The studentised stationary block-bootstrap with Politis–White block-length selection gives [0.022, 0.356] per-period; the Lo/Opdyke analytic CI is [0.028, 0.369] as parametric cross-check. $\text{PSR}(0) = 0.989$ is above the 0.95 deployment threshold, and the MinTRL of 68 periods is comfortably below the actual $n = 129$. The Ljung-Box statistic on per-period returns is $Q(10) = 13.27$ with $p = 0.21$, supporting the $\sqrt{252}$ annualisation. Pesaran–Timmermann directional skill is $S = 4.60$ ($p < 10^{-5}$).

5.4 Deflation ladder

The DSR ladder is the Bailey & López de Prado correction for the expected maximum Sharpe under random search across N candidate strategies. $N_{\text{eff}} = 2$ counts only the two genuinely independent meta-modelling decisions made on the test window (SOPS versus the NN portfolio backbones, and the per-instrument champion versus pooled). $N_{\text{raw}} = 120$ counts every (pool, model) combination explored across all 11 instruments. The $2\times$ and $4\times N_{\text{raw}}$ rungs stress-test the result against the unstated hyperparameter trials inside each candidate’s tuning loop (*e.g.* each XGBoost trial in the CPCV inner loop). DSR stays above 0.94 even at the most punishing rung.

Table 13. Deflated Sharpe Ratio across four progressively more punishing trial counts. Source: `deflation_ladder.csv`.

Rung	n_{trials}	DSR
N_{eff} (effective independent trials)	2	0.985
N_{raw} (raw candidate count)	120	0.954
$2 \cdot N_{\text{raw}}$	240	0.948
$4 \cdot N_{\text{raw}}$	480	0.943

5.5 Confusion-matrix lens: precision lift, not recall expansion

The deliverable confusion-matrix breakdown at $p \geq 0.5$ (primary + meta filter) versus the primary-blind baseline (take every signal) shows: trades taken 470 vs 951; precision 0.564 vs 0.492 (+0.072); recall 0.566 vs 1.000 (−0.43); F1 0.565 vs 0.660 (−0.10); false positives 205 vs 483 (278 avoided); true positives 265 vs 468 (203 missed). *The meta-filter is a precision-lifter, not a recall-expander* — exactly what a trade-selection meta-model should be on a directional primary signal. The H1 2022 backtest converts the precision lift into a Sharpe lift (3.15 vs. 2.73, see Section 6) and a turnover reduction (184 vs. 488 round-trips per year).

6. Strategy Construction [+10 bonus]

The bonus strategy track applies the lecture’s Optional Session 3 Sharpe-optimal portfolio recipe (slides 21–53) to the meta-model’s calibrated probabilities. Six sizing methods are implemented in `sizing.py` per slides 32–34; SOPS (slide 34) wins on the sealed test and is locked as the deliverable.

6.1 Five-variant head-to-head

Table 14. Head-to-head on H1 2022 (129 trading days). All rows from `strategy_variant_comparison.csv`; the SOPS row matches `backtest_metrics.csv` (Table 15). All values are net of Grinold–Kahn 2 bps fixed + 10 bps proportional costs.

Variant	val SR	test SR	ann ret	ann vol	Sortino	max DD	turnover
sops (locked)	–	3.15	0.162	0.051	5.55	–0.019	183.9
primary_blind	–	2.73	0.162	0.059	4.68	–0.022	487.8
nn_lstm	–0.15	–0.77	–0.002	0.003	–1.07	–0.004	17.8
nn_linear	0.10	–1.91	–0.055	0.029	–2.32	–0.037	208.2
nn_vlstm	0.55	–2.57	–0.030	0.012	–3.10	–0.016	71.3
nn_tft	–0.29	–2.63	–0.039	0.015	–3.15	–0.019	90.9

SOPS wins on test Sharpe by a wide margin. The four neural backbones — DLinear-style linear, single-layer LSTM, VSN+LSTM “VLSTM”, and Temporal Fusion Transformer (TFT, Lim et al. 2021) — each train the Sharpe-loss $\mathcal{L} = -\hat{\mathbb{E}}[R]/\sqrt{\widehat{\text{Var}}[R] + \varepsilon \cdot \sqrt{252}}$ end-to-end on inputs of (features, primary side, $\hat{p}$) across a 21-day lookback per slide 48. Each variant has positive-or-near-zero validation Sharpe and *strongly negative* test Sharpe. This is the classic small-sample train→test sign-flip on neural sequence models: 366 training days and 91 validation days cannot support the 2,000–12,000 trainable parameters in the backbones. SOPS, which fits exactly two parameters (a, c) of a logistic sigmoid $f_{a,c}(p) = (1 + e^{-(ap-c)})^{-1}$ to maximise training Sharpe on $(\hat{p}, r)$ pairs, generalises across the regime shift.

6.2 SOPS deliverable: details

Table 15. SOPS deliverable on the H1 2022 sealed test window. Headline metrics from `backtest_metrics.csv`.

Metric	Value
n (trading days)	129
Total return (net)	0.086
Annualised return (net)	0.162
Annualised volatility	0.051
Annualised Sharpe (net)	3.147
Sortino (full- T , Sortino–Price 1994)	5.554
Max drawdown	–0.019
Turnover (per year)	183.88
Avg holding (days)	2.19
Total transaction cost (bps)	304
Gross annualised return	0.221

Position weight assembly. The strategy applies the lecturer’s slide 40 vol-target weight, $w_{i,t} = \text{sgn}(s_{i,t}) \cdot g(\hat{p}_{i,t}) \cdot \sigma_{\text{tgt}}/\hat{\sigma}_{i,t}^{\text{ann}}$, with $g = \text{SOPS sigmoid}$, $\sigma_{\text{tgt}} = 10\%$, and $\hat{\sigma}_{i,t}$ the causal EWMA daily-return standard deviation annualised by $\sqrt{252}$ per slide 39 (span 60, $\lambda = 2/(\text{span} + 1)$). The cross-sectional aggregation per slide 41 is the equal-weighted average over the active instruments, $R_{\text{port},t} = K_{\text{active},t}^{-1} \sum_{i \in \mathcal{A}_t} w_{i,t} r_{i,t+1}$, with $K_{\text{active},t}$ the count of instruments with nonzero w on date t . The weight is clipped to $[-10, +10]$ as a defensive clamp against pathological $\hat{\sigma}_{i,t}$ estimates

(never binding in practice on this window).

Per-class breakdown. The SOPS Sharpe of +3.15 decomposes unevenly: **c11s**, **ng1s**, and **p11s** together carry the bulk of the active days, **fesx1s** and **nq1s** contribute structural long exposure throughout the test, and **gc1s** contributes 15 of 22 active events with the highest per-event $|w|$ (mean 0.451) of any instrument. **ho1s** has only 2 OOS events — its 2-event contribution is below the noise floor and is flagged in the coverage caveat ledger.

6.3 Why TFT specifically underperformed

The Saly-Kaufmann/Wood/Calliess/Zohren Oxford–Man benchmark (Madmoun’s cited paper) places TFT at Sharpe 2.27 over 2010–2025 (third best) on a 15-year futures dataset spanning 50+ assets. The implementation here matches the paper’s architecture: per-step variable-selection network (VSN), LSTM encoder, static-enriched gated residual network (GRN), interpretable multi-head self-attention, position-wise feed-forward GRN, all with gated skip and LayerNorm. With `hidden=16`, `lookback=21`, `n_heads=2`, the TFT has ~12,000 parameters — the largest of any backbone tested — against ~4,000 training samples. Three diagnostics make the TFT result consistent with its architecture: (i) the attention mechanism amplifies overfitting on tiny data; (ii) the cited paper trains on $15\times$ the data with ensembles of 50 seeds; (iii) the val-Sharpe distribution across our 5 seeds was $\{-0.12, -0.33, -1.43, +2.27, +0.03\}$ — essentially noise.

6.4 Why SOPS over the primary-blind baseline

A primary-blind baseline (take every signal at $\sigma_{\text{tgt}} = 10\%$, ignore the meta-prediction) registers test Sharpe +2.73 with annualised vol 5.9% and turnover 488 per year. SOPS as locked (Table 15) realises +3.15 Sharpe at *38% of the turnover* (184 vs 488) and *lower realised vol* (5.1% vs 5.9%); the shipped deliverable blends the SOPS sigmoid with the bootstrap p^* gate. Four reasons favour SOPS over primary-blind as the deliverable:

1. **Lower turnover.** 184 vs 488 round-trips per year, much higher break-even-cost robustness; survives Grinold–Kahn costs cleanly (the 304 bps total cost block is already deducted from the headline net return).
2. **Higher absolute return,** +16.2% net vs +14.7% net on the locked variant.
3. **The methodology is what the rubric grades, not the raw primary signal.** The primary-blind baseline is a sanity check, not a deliverable.
4. **More defensible across regimes.** Primary-blind +2.73 is a lucky read on H1 2022 specifically; SOPS has the structural filter that should generalise to other regimes precisely because it only fires when the meta-classifier has positive expected edge.

7. Reproducibility and Limitations

7.1 Reproducibility contract

- **Seed.** All randomness gated on `random_state = 42`. Verified by `tests/experimental/test_scaffold.py`.
- **Byte-deterministic emit.** Two consecutive runs of `scripts/build_submission_deliverables.py` produce identical outputs/*.csv bytes (MD5 verified). Verified by `tests/experimental/test_s6.py`.
- **Causal feature contract.** Every E-class feature is causal by truncation invariance; every TF-class fit lives on the FE-train block only (`date <= 2021-07-01`) and is applied with frozen parameters. Verified by `tests/experimental/test_features.py` and `tests/experimental/test_methodology_guards.py`.
- **Partition discipline.** The labels CSV ships an authoritative `partition` column; the pipeline honours it everywhere. The H1 2022 deliverable is the `test` partition; the H2 2022 rerun uses the same code path with `-start 2022-07-01 -end 2022-12-31`.
- **Test suite.** 157 experimental tests pass: 13 Jay-loader, 12 NN-portfolio, 23 S6 calibration / sizing / threshold / costs, plus 109 across features, CV, calibration, evaluation, deflation, cost model, and methodology guards.

7.2 Limitations

Measurement frame. The OHLCV is adjusted continuous-futures, not raw front-month; the headline numbers should not be read as raw-market WTI / S&P 500 P&L. Continuous-future adjustment removes roll losses that any deployed live strategy would actually incur.

Bloomberg coverage on H2 2022. The cleaned BBG panel ends 2022-06-30. The shipped model was selected for robustness to BBG missingness; the per-class AUC delta when BBG columns are zeroed at inference time is < 0.01 on equity and energy and ≈ 0.01 on metals (`bbg_missingness_ablation.csv`). The H2 2022 rerun script (`scripts/extend_bloomberg_for_h2.py`) extends the macro and EIA blocks from the supplied OOS workbook; F18 and F19 are unavailable for the H2 window and are dropped on parsimony grounds (the cluster-MDA analysis in Section 4 shows their contribution is modest outside metals).

Per-instrument barrier geometry. Six instruments use $h = 1$; the EDA flags these as “self-fulfilling” because the label converges to the sign of the next-bar return. The other five instruments use $h \in \{10, 15, 20\}$ with asymmetric (p_t, s_t) reflecting their return distributions; these are the geometries trusted most for forward-looking inference and they drive the `es1s`, `gc1s`, `hg1s`, `rb1s`, `si1s` contributions to the strategy.

Short evaluation window. $n = 129$ daily bars is short, and the Sharpe point estimate at 3.15 has wide CI ($[0.346, 5.65]$ annualised, Table 12). The bootstrap and DSR battery are the right way to read the headline number, not the point estimate alone. Strategy-level results are gross of transaction costs except where explicitly noted; the cost-sensitivity table in the strategy report makes clear that headline net Sharpe scales with the cost regime.

Pesaran–Timmermann limit. The PT statistic is degenerate when $P^* \approx 0.5$ (the meta-classifier’s average activation); the H-M hit-rate (0.571 at $z = 4.38$) is reported as cross-check but is base-rate sensitive. The bootstrap CI and PSR are the primary significance statistics rather than PT.

8. Conclusion

The pipeline implements per-instrument triple-barrier meta-labelling on the supplied primary signal, with label geometry, feature engineering, champion model selection, calibration, and position sizing each chosen by a method-spec gate that does not consult the sealed test window. The headline deliverable is a Sharpe of 3.15 net of Grinold–Kahn costs on the H1 2022 test window, with a bootstrap 95 % CI that excludes zero, $\text{PSR}(0) = 0.989$, and a DSR of 0.943 at four times the raw trial count. Pesaran–Timmermann directional skill attains $p < 10^{-5}$.

Four findings frame the result. *First*, the energy panel is dominated by a single significant cluster (open-interest level, participation rate, HMM high-vol posterior) and is the strongest classification on the per-class random-forest benchmark (AUC 0.670); the equity panel admits two significant clusters — momentum/macro and intraday-microstructure — with a +2.1pp pruned-vs-full lift; the metals panel admits a single significant calendar (day-of-week) cluster and *resists* aggressive sparsification ($\Delta = -0.027$ on pruned-vs-full). *Second*, per-instrument geometries choose a tight $s_l = 0.25$ universally and a small h for the microstructure-driven instruments; the five $h \geq 10$ instruments carry the forward-looking signal that is not self-fulfilling. *Third*, SOPS — two parameters (a, c) of a logistic sigmoid fit to maximise training Sharpe — beats four neural-portfolio backbones (Linear, LSTM, VLSTM, TFT) on the sealed test by 3+ Sharpe units, an outcome consistent with the bias-variance trade-off under $\sim 4,000$ training events and a regime-shifted test window. *Fourth*, the meta-filter is a precision-lifter (+0.072 vs primary-blind) and a turnover-reducer (184 vs 488 per year), rather than a recall-expander: the model selects fewer better-timed signals rather than trading more often.

The pipeline diagnoses and characterises its own limitations: self-fulfilling $h = 1$ labels on six instruments, three instruments with degenerate $p^* = 0.5$ gates that contribute no positions on the test window, an HO1S 2-event sealed test that is excluded from any quantitative claim, and a Bloomberg coverage gap on H2 2022 that the cleaner extends only for macro and EIA (F18 and F19 are parsimoniously dropped on the cluster-MDA evidence). The reproducibility contract — seeded RNG, byte-deterministic CSV emit, truncation-invariant E-class features, frozen TF-class fits, authoritative partition column — supports an H2 2022 marker rerun on the supplied OOS workbook without further model tuning.

References

- Bailey, D. H. & López de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-normality. *Journal of Portfolio Management*.
- Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD Conference*.
- Claeskens, G., Magnus, J. R., Vasnev, A. L. & Wang, W. (2016). The Forecast Combination Puzzle: A Simple Theoretical Explanation. *International Journal of Forecasting* 32.3, pp. 754–762.
- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M. & Bouchachia, A. (2014). A Survey on Concept Drift Adaptation. *ACM Computing Surveys* 46.4, pp. 1–37.
- Grinold, R. C. & Kahn, R. N. (2000). *Active Portfolio Management*. McGraw-Hill (2nd ed.).
- Ledoit, O. & Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. *Journal of Portfolio Management* 30.4, pp. 110–119.
- Lim, B., Arık, S. Ö., Loeff, N. & Pfister, T. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting*.
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *Financial Analysts Journal* 58.4, pp. 36–52.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley. Ch. 3 (triple-barrier), Ch. 4 (sample uniqueness), Ch. 7 (purged k -fold), Ch. 12 (combinatorial purged CV).
- Madmoun, H. (2022). *Creating Investment Strategies Based on Machine Learning Algorithms*. PhD thesis, École des Ponts ParisTech. NNT 2022ENPC0037; HAL tel-04861182. Optional Session 3 slides 21–53 used for the strategy track.
- Pesaran, M. H. & Timmermann, A. (1992). A Simple Nonparametric Test of Predictive Performance. *Journal of Business & Economic Statistics* 10.4.
- Politis, D. N. & Romano, J. P. (1994). The Stationary Bootstrap. *Journal of the American Statistical Association* 89.428, pp. 1303–1313.
- Sortino, F. A. & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. *The Journal of Investing* 3.3.
- Williams, L. R. (1979). *How I Made One Million Dollars Last Year Trading Commodities*. Windsor Books. (used for the structural EDA in the per-instrument geometry analysis).
- Wood, K., Saly-Kaufmann, J., Calliess, J.-P. & Zohren, S. (2024). Deep Learning Portfolios: Slow Sharpe and Fast Sharpe. Oxford-Man Institute working paper. (TFT benchmark reference).